Mohammad Refat

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Research Interests

Computational astrophysics, Exoplanets, Time-series analysis, Inverse problems, Statistical inference, Scientific computing

Education

CUNY Graduate Center 2023–2025
M.S. in Astrophysics
Thesis: *Starspot Inference Using Light Curve Inversion Techniques*

CUNY Baccalaureate for Unique and Interdisciplinary Studies 2019–2022
B.S. in Computational Astrophysics, *cum laude*

Bernard M. Baruch College 2017–2019

Research Experience

Starspot Inference with Light Curve Inversion Techniques Aug 2023 – Sep 2025
American Museum of Natural History
Advisor: Dr. Lucy Lu

- Developed computational approaches for starspot inference using light curve inversion techniques.
- Applied time-series analysis and probabilistic modeling to characterize stellar surface features.
- Implemented Python-based scientific computing workflows for astrophysical inference and model evaluation.

Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres May 2021 – Aug 2023
American Museum of Natural History
Advisor: Dr. Johanna Vos

- Investigated atmospheric variability in brown dwarfs and giant exoplanets through analysis of photometric light curves.
- Developed computational workflows for mapping cloud structures from observational time-series data.
- Applied quantitative modeling techniques to infer atmospheric properties from astronomical observations.

Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2 May 2020 – May 2021
Sloan Digital Sky Survey Faculty and Student Team
Advisor: Dr. Allyson Sheffield

- Analyzed APOGEE-2 spectroscopic data to identify candidate stellar members of the Jhelum stellar stream.
- Applied statistical and computational analysis methods to characterize stellar populations.
- Contributed to a refereed publication in *The Astrophysical Journal*.

Looking at Star Formation Through Chemistry Aug 2018 – May 2020
American Museum of Natural History
Advisor: Dr. Allyson Sheffield

– Analyzed stellar spectra to derive chemical abundances and characterize stellar populations.
– Processed and normalized echelle spectra and measured equivalent widths of spectral lines.
– Applied quantitative techniques to spectroscopic datasets.

Two-Point Statistics in the Star Forming ISM Jun 2018 – Aug 2018
Center for Computational Astrophysics, Flatiron Institute
Advisor: Dr. Chang-Goo Kim
– Performed statistical analysis of metallicity correlations in star-forming regions.
– Investigated relationships among stellar metallicity, velocity, and star formation rate.

Refereed Publications

Sheffield, Allyson A. et al. (May 2021). "Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2". In: *The Astrophysical Journal* 913.1, p. 39. ISSN: 1538-4357. DOI: 10.3847/1538-4357/abee93. URL: http://dx.doi.org/10.3847/1538-4357/abee93.

Academic Presentations

Talks

"Starspot Inference with Light Curve Inversion Techniques" 2025
CUNY Masters Graduation 2025
Simons Foundation

"Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres" 2022
240th American Astronomical Society Meeting
American Astronomical Society

"Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres" 2021
Center for Computational Astrophysics / City University of New York / American Museum of Natural History Symposium
Center for Computational Astrophysics

"Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2" 2020
American Museum of Natural History REU Symposium
American Museum of Natural History

"Where Were Stars Born in the Milky Way?" 2019
American Museum of Natural History REU Symposium
American Museum of Natural History

"Two-Point Statistics in the Star Forming ISM" 2018
Center for Computational Astrophysics REU Symposium
Simons Foundation

Posters

"Starspot Inference with Light Curve Inversion Techniques" 2025
245th American Astronomical Society Meeting
American Astronomical Society

"Star Spots with Starry" 2024
22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun
University of California San Diego

"Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres" 2023
Center for Computational Astrophysics Gotham Fest 2023
Center for Computational Astrophysics

"Towards Mapping Brown Dwarf and Giant Exoplanet Atmospheres" 2023
241st American Astronomical Society Meeting
American Astronomical Society

"Chemodynamically Characterizing the Jhelum Stellar Stream with APOGEE-2" 2021
237th American Astronomical Society Meeting
American Astronomical Society

Teaching

Adjunct Lecturer Aug 2025 – May 2026
Baruch College
- Introductory Physics Lecture
- Introductory Physics Laboratory
- Physics II Laboratory

Technical Skills

Programming	Python, MATLAB, Bash
Scientific Python	NumPy, SciPy, Pandas, Matplotlib, Jupyter Notebook, Astropy, Lightkurve, Astroquery
Computational Methods	Time-Series Analysis, Statistical Modeling, Inverse Problems, Bayesian Inference, Computational Modeling, Spectral Analysis
Software	Git, Linux, LaTeX, Scientific Writing